

IBM HR Analytics — Predict Employee Attrition

Elevate Labs Internship • Data Analytics • May 2025 • Tools: Python | Sklearn | SHAP | Power BI

Abstract

IBM Corporation faces a 16.12% employee attrition rate, with Sales (20.63%) and HR (19.05%) departments most affected — costing an estimated 6–9 months salary per resignation. Using the IBM HR Analytics dataset (1,470 employees, 35 features), this project identifies root causes of attrition and builds a Logistic Regression model with SMOTE oversampling, achieving 87% accuracy and 54% recall. The model enables HR managers to proactively identify at-risk employees and implement targeted retention strategies before resignations occur.

Introduction

Employee attrition disrupts productivity, increases hiring costs, and causes knowledge loss. This project answers: "Which employees are most likely to leave IBM — and why?" Through exploratory data analysis and machine learning, the project provides HR leadership with actionable insights to design targeted retention programs, optimize compensation, and improve workplace conditions before resignations occur.

Tools Used

Tool / Library	Purpose
Python + Pandas	Data cleaning, transformation and analysis
Seaborn + Matplotlib	EDA visualizations — countplots, boxplots, heatmaps
Scikit-learn + SMOTE	ML models, StandardScaler, confusion matrix, metrics
SHAP	Explainable AI — global and individual prediction explanations
Power BI	Interactive HR analytics dashboard with 2 pages
Streamlit	Web application for live attrition prediction
IBM HR Dataset (Kaggle)	1,470 employee records, 35 features

Steps Involved in Building the Project

1. Data Understanding & Problem Statement:

Loaded IBM HR dataset and confirmed 16.12% overall attrition rate. Department-wise analysis revealed Sales (20.63%) and HR (19.05%) as most affected. A data-backed business problem statement was formulated before modelling.

2. Data Cleaning & Preprocessing:

Dropped constant columns (EmployeeCount, StandardHours, Over18) and EmployeeNumber. Applied label encoding (Attrition, OverTime, Gender), ordinal encoding (BusinessTravel: 0/1/2), one-hot encoding with drop_first=True (Department, JobRole, EducationField, MaritalStatus), and StandardScaler on 14 continuous columns. Final dataset: 1,470 rows × 44 features.

3. Exploratory Data Analysis (EDA):

Five business-driven charts: (i) Overtime — employees working overtime are 4x more likely to leave (31% vs 10%). (ii) Salary Boxplot — leavers earned ₹1,600/month less (₹3,700 vs ₹5,300 median). (iii) Age Histogram — 20–25 age group shows 36% attrition. (iv) Work Life Balance — bad WLB group has 31% attrition. (v) Correlation Heatmap — overtime strongest positive driver; TotalWorkingYears strongest protective factor.

4. Machine Learning Model:

Model	Accuracy	Recall	Selected?
Logistic Regression + SMOTE	87%	54%	✓ Yes — Best Recall
SVM + SMOTE	88%	51%	✗ Lower Recall
Random Forest + SMOTE	88%	28%	✗ Poor Recall
Decision Tree + SMOTE	77%	44%	✗ Lower overall

Data split 80/20. SMOTE applied to training data only to handle class imbalance (84% stayed vs 16% left). Logistic Regression selected for highest Recall — most critical metric since missing an at-risk employee is far more costly than a false alarm.

5. SHAP Analysis:

SHAP LinearExplainer identified global feature importance and explained individual predictions. Key finding: Overtime is the top attrition driver; JobLevel and TotalWorkingYears are top retention factors. Waterfall plots generated per employee to explain individual risk scores.

6. Power BI Dashboard + Streamlit App:

Two-page Power BI dashboard: Page 1 — IBM-themed dark dashboard with 5 charts (KPI cards, department attrition, overtime impact, salary bands, age group). Page 2 — Key Findings page with 3 pastel recommendation boxes. A Streamlit web app was also deployed for real-time HR prediction with personalized suggestions based on top risk factors.

Conclusion

Three primary attrition drivers identified: overtime burden (31% vs 10%), salary gap (₹1,600/month), and young employee career exploration (36% attrition in 20–25 group). Recommended HR interventions: overtime scheduling controls, quarterly compensation benchmarking, and mentorship programs for young employees. The model (87% accuracy, 54% recall) enables HR to identify at-risk employees before resignation — converting reactive damage control into proactive talent retention.